Moderate Exercise on Voluntary Apnea Duration in Healthy Adult Male Across a Spectrum of Body Mass Index

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ABSTRACT

The cessation of breathing is called apnea or breath-holding. This study aimed to investigate the impact of moderate exercise on voluntary apnea duration in healthy adult males across a spectrum of Body Mass Index (BMI), to elucidate the physiological interplay between adiposity and respiratory-metabolic reserve. Thirty male subjects (aged 20-30 years) were stratified into normal weight (n=10, BMI: 22.42±0.41 kg/m²), overweight (n=10, BMI: 27.47±0.45 kg/m²), and obese (n=10, BMI: 37.46±0.97 kg/m²) groups. The duration of voluntary apnea was measured at rest and immediately following a session of moderate exercise. Data were presented as Mean ± SEM and were compared using one-way ANOVA followed by Tukey's post-hoc text at $p<0.05$. A significant decrease in apnea duration was seen after exercise compared to before exercise in all the groups. Also, there was a significant decreased in apnea duration in the overweight and obese groups when compared to the normal weight group. These findings demonstrate an adiposity dependent impairment of voluntary apnea capacity with increasing BMI, an effect markedly exacerbated by the metabolic stress of exercise. The results suggest that elevated adiposity, particularly obesity, critically erodes integrated cardiorespiratory and metabolic reserve, limiting the tolerance to respiratory challenges.

Keywords: Voluntary apnea, Body mass index, Obesity, Exercise, Respiratory physiology.

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1. INTRODUCTION

The voluntary cessation of gas exchange between the lungs and the atmosphere is referred to as voluntary apnea or breath holding [1]. Apnea can also be involuntary if it is induced by means other than the will of the individual, such as by drugs or neurological disorders [2, 3]. During breath-holding, carbon dioxide levels can be significantly raised in the lungs and blood to a "breaking point". At this point, the body's instinct to breathe takes over, ensuring that the body gets enough oxygen to prevent unconsciousness [4]. This involuntary respiratory act overrides voluntary breath-holding and helps to maintain the body's oxygen levels [5, 6]. The average breath-holding duration (BHD), according to researchers, appeared to be a mean value of 78 seconds when holding the breath at total lung capacity (TLC) with a variable range of 72-78 seconds [7]. Breath-holding duration can vary significantly based on mechanical factors like lung volume, chemical factors like pCO_2, pO_2 and H^+ ion concentration, non-chemical factors like involuntary muscular contractions, psychological factors like motivation, stress, competition, extrinsic factors like training [8] and muscular exercise.

The ability to suppress respiratory urges and sustain breath-holds for prolonged durations relies on three fundamental principles: the capacity for oxygen storage (lungs, blood (haemoglobin) and skeletal muscle (myoglobin), the efficacy of oxygen

conservation and utilisation (in large part dependent on cardiovascular and metabolic adjustments), and volitional factors (psychological tolerance to the increasing urge to breathe and the continuously intensified IBM) [9].

The current metric for defining anthropometric height/weight features and for categorizing them into groups is known as the body mass index (BMI). The common interpretation is that it represents an index of an individual's fatness [10]. In a study on the effect of obesity on peripheral chemoreflex sensitivity in voluntary apnea, a ventilatory response to hypoxia was observed, such as a strong inverse correlation between the sensitivity of the peripheral chemoreflex to CO_2 and breath-holding duration in healthy men [11]. Furthermore, they explained that both the apnea duration and peripheral chemoreflex sensitivity to Co2 were markedly reduced in subjects with high BMI than those with normal BMI. Based on this, they concluded that the breath-holding duration is not affected by the weight or BMI, but by the body composition, which reflects not only the effect of obesity on the respiratory function but also its effect on metabolism and gas exchange [11]. While the independent effects of obesity on pulmonary mechanics and chemoreflex sensitivity have are well documented, and the ventilatory response to exercise is similarly established, there remains a paucity of integrative research examining how these factors interact dynamically. Therefore, this study aims to evaluate effects of moderate exercise on the duration of voluntary apnea in male subjects with varied body mass index (overweight and obese).

2. MATERIALS AND METHODS

2.1. Subjects: Thirty male subjects within the age range of 20-30 years old were used for this study as they were divided into normal, overweight and obese groups with ten (10) subjects per group. The study enrolled exclusively healthy male participants. Inclusion criteria include individuals free from tobacco use, chronic or acute medical conditions, and not on any medications. Individuals with a history of alcohol or substance dependence, or a diagnosis of obstructive sleep apnea and regular practice of breath-hold were excluded. Participants were assessed as being in good general health, engaged in only moderate physical activity, and were not consumers of nicotine containing products. Prior to enrollment, all participants were provided with detailed information regarding the study protocol and associated potential risks. Written informed consent was obtained from each individual, and the study protocol received formal approval from the Research and Ethics Committee of the University of Benin.

2.2. Experimental design: All thirty adult male subjects were divided into normal, overweight and obese groups with ten (10) subjects per group. All subjects were asked to stay away from caffeine, tobacco and alcohol for 12 hours or more ahead of the study. Anthropometric characteristics such as height and weight were measured using standard methods [12]. Metric units: weight (kilograms) divided by height squared (meters) [10]

$BMI = kg/m^2$

In all subjects, breath-holding test was performed before breakfast.

2.3. Experimental procedures: The voluntary apnea test was conducted under controlled environmental conditions, with the room temperature maintained at 25°C. For each measurement, the participant was directed to take a voluntary inspiration and then hold their breath. The apnea duration was recorded from the onset of this inspiration until the point of breath-hold termination. This procedure was repeated to obtain three separate measurements, and the final reported value represented the mean duration calculated from these trials.

Participants performed a moderate-intensity exercise session on a mptorixed treadmill for

the post-exercise apnea assessment. The exercixe consisted of brisk walking at a speed of 5-6km/h for 10 minutes, with intensity maintained at approximately 50-70% of the age-predicted maximum heart rate. Following exercise, participants ressted for one minute before the post-exercise voluntary apnea test was conducted as before exercise.

2.4. Statistical analysis: Data was presented as Mean ± SEM and compared using one-way ANOVA followed by Tukey's post-hoc text at $p<0.05$ of Graph pad prism version 8.1 statistical software.

3. RESULTS

Table 1 shows the different categories of body mass index of the subjects. Figure 1 showed bar charts showing voluntary apnea duration before and after exercise in different body mass index. Duration of apnea decreased after the exercise compared to before exercise in all the groups. There was a significant decrease ($p<0.05$) in duration of apnea in overweight (before exercise 34.70±0.86 secs and after exrcise 19.30±0.58 secs) and obese (before exercise 28.70±1.27 secs and after exercise 12.60±0.62 secs) as compared with normal weight (before exercise 45.00±2.39 secs and after exercise 23.40±1.51 secs).

Table 1. Comparing the different categories of body mass index

	Normal Weight	Overweight	Obese	*P*-value
Mean BMI (Kg/m^2)	22.42 ± 0.41	27.47 ± 0.45*	37.46 ±0.97*	0.0001

Values are represented as Mean ± SEM; n =10, * Superscript indicates significant difference of all treatment group before and after exercise.

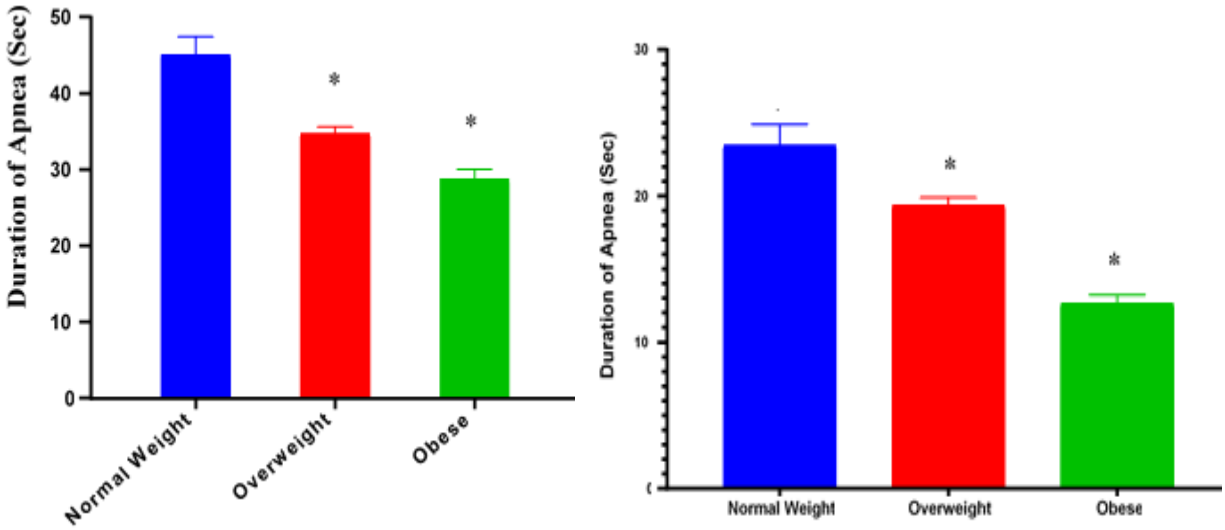


Figure 1. Comparing the mean values of the voluntary apnea duration before (a) and after exercise (b) on the different categories of body mass index

**p< 0.05 indicates significant difference, when overweight and obese are compared with normal weight.*

4. DISCUSSION

Voluntary apnea, the deliberate cessation of breathing, serves as a potent stress test for integrated respiratory and metabolic function. Figure 1a presents a clear and significant inverse relationship between BMI and the duration of voluntary apnea before exercise, establishing a graded physiological impairment across adiposity categories. There was a

significant and graded decrease in the duration of voluntary apnea in the overweight and obese groups, compared to the normal-weight group. This pre-exercise baseline disparity may be a manifestation of underlying, chronic physiological adaptations and burdens imposed by excess adiposity. This observed gradient in pre-exercise apnea capacity can be attributed to at least three interlinked physiological mechanisms, each exacerbated by increasing BMI. Firstly, excess adipose tissue, particularly in a central or visceral distribution may act as a mechanical load on the respiratory apparatus. Already established is the fact that adiposity in the chest wall, abdomen, and thoracic cavity reduces chest wall compliance, which can impede diaphragmatic descent, and most critically, decrease functional residual capacity (FRC) and expiratory reserve volume (ERV) [13].

FRC represents the lung volume at the end of a normal exhalation and constitutes a primary reservoir for oxygen during breath-holding. A reduction in FRC, a well-documented consequence of increasing BMI, directly diminishes oxygen stores [14]. Consequently, during voluntary apnea, individuals with higher BMI may experience a more rapid decline in arterial oxygen partial pressure (PaO_2), reaching the hypoxic threshold that triggers the involuntary breakpoint of apnea sooner than those with greater lung volumes. This opinion corroborated a study by Jones and Nzekwu [14] who demonstrated an inverse correlation between BMI and static lung volumes, providing a direct structural basis for the findings in this study. Secondly, this may also be as a result of the proinflammatory cytokines from the adipose tissues in obese and overweight individuals, stimulating the peripheral and central chemoreceptors to drive inhalation. It is known that the adipose tissue is an active endocrine organ, capable of secreting leptin, which is an adipokines capable of low-grade inflammation [15]. As leptin is a stimulant of ventilation, these pro-inflammatory cytokines can potentiate chemoreceptor responses to hypoxia and hypercapnia. Such that CO_2 accumulation or O_2 depletion during apnea, can cause a stronger respiratory drive capable of terminating breath-holding in individuals with overweight and obesity, leading to a shorter duration of apnea [16].

Also, leptin may act as a potent respiratory stimulant at the level of the central nervous system [17]. Therefore, hyperleptinemia, a state that is common in obesity, may heighten the sensitivity of central chemoreceptors to hypercapnia (elevated $PaCO_2$), resulting in a need to breath. Barnes et al., supports the association between obesity, hyperleptinemia, and an elevated hypercapnic ventilatory response, offering a neurohumoral explanation for the reduced tolerance to CO_2 accumulation during apnea. In their study, they stated that Leptin may act in neurones that are part of the neural circuitry involved in chemoreflex, such as those located in the Bötzinger complex in which leptin receptors are present [18]. In contrast, [19] showed in their study that hyperleptinaemia is associated with a reduction in respiratory drive and hypercapnic response, suggesting an extension of leptin resistance to the respiratory centre. However, studies by health economists like Burkhauser and Cawley have argued that since BMI is a crude metric that fails to distinguish between adiposity and lean muscle mass, an athlete with a high BMI due to muscularity would possess lung volumes and autonomic profiles starkly different from an individual with adiposity-driven obesity [20]. Which may mean that increased or high BMI may not necessarily

translate into a shortened voluntary breath holding time.

Figure 1b represents the duration of voluntary apnea following moderate exercise across a gradient of BMI. The data revealed a significant and graded decline in the duration of apnea through both the overweight and obese individuals respectively compared to the normal weight. This post-exercise state is a replication of pre-exercise trends, and also represents a physiological "stress test" that may have resulted from a disproportionately challenged system already burdened by adiposity. The universal reduction in apnea time confirms the expected metabolic perturbation induced by exercise. The shortened apnea times across all groups post-exercise and the gradient of impairment may be explained by the combined effect of the high sensitization caused by hyperleptinemia and increased hydrogen ion concentration on the central chemoreceptors.

Exercise increases CO_2 production and often induces a transient metabolic acidosis via lactate accumulation [21]. The drive to breathe post-exercise is powerfully stimulated by elevated arterial hydrogen ion concentration, which acts primarily on peripheral chemoreceptors, and by hypercapnia, sensed by central chemoreceptors [22]. Individuals with obesity are postulated to have a pre-sensitized chemoreflex axis due to chronic hyperleptinemia and inflammation [17]. Thererefore, this double sensitization effect means the same post-exercise elevation in $PaCO_2$ and H^+ generates a disproportionately stronger respiratory drive, making the conscious suppression of breathing (apnea) intolerable much sooner than in individuals with normal chemosensitivity. Previous studies confirm an exaggerated ventilatory response to exercise in obesity, linked to both increased metabolic load and chemosensitivity [22].

Also, obesity has been implicated to cause a reduced FRC, resulting in a diminished reservoir of oxygen at the start of apnea [13]. Conversely in post-exercise, the metabolic rate remains elevated, increasing the rate of oxygen consumption ($\dot{V}O_2$). Consequently, in individuals with obesity, a smaller oxygen store may be consumed at a faster rate, leading to a precipitous drop in arterial oxygen saturation (SpO_2). Since the breaking point of voluntary apnea is closely linked to the onset of critical hypoxia, this combination of low storage and high consumption may catastrophically shorten tolerable breath-hold duration [23].

CONCLUSION

This study demonstrates a clear, dose-dependent relationship between increasing BMI and a reduction in voluntary apnea tolerance, an impairment that is profoundly exacerbated by the metabolic stress of moderate exercise. The findings reveal that individuals with overweight and, more severely, obesity, exhibit a significantly diminished capacity to suppress the drive to breathe, both at rest and following physical exertion.

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